MTH309 PRACTICE PROBLEMS WEEK 2

Question 1. Show that the following collections of sets generate the Borel σ-field $\mathbb{B}_{\mathbb{R}}$ on $\mathbb{R}$.

(a) All compact sets in $\mathbb{R}$.
(b) All intervals of the form $(-\infty, a), a \in \mathbb{R}$
(c) All intervals of the form $(-\infty, a], a \in \mathbb{R}$
(d) All intervals of the form $(a, \infty), a \in \mathbb{R}$
(e) All intervals of the form $[a, \infty), a \in \mathbb{R}$

Question 2. Recall the collection of finite disjoint unions of left-open right-closed intervals in $\mathbb{R}$, which was discussed in the class. Verify that this is a field.

Question 3. Recall the Dirac measure and counting measure discussed in the class. Write down an argument showing the countable additivity property for these set functions.

Question 4. Let $\{\mu_n\}_n$ be a sequence of measures defined on a measurable space $(\Omega, \mathcal{F})$. Let $\{\alpha_n\}_n$ be a sequence of non-negative real numbers. Show that the set function $\nu := \sum_{n=1}^{\infty} \alpha_n \mu_n$ is a measure.

Question 5. Let Ω be a finite or countably infinite set. Take $\mathcal{F} = 2^{\Omega}$, the power set of Ω. Show that all probability measures $\mathbb{P}$ on $(\Omega, \mathcal{F})$ is given by $\mathbb{P} = \sum_{\omega \in \Omega} p_\omega \delta_\omega$, where $p_\omega := \mathbb{P}(\{\omega\}), \forall \omega \in \Omega$.

Question 6. Complete the proof of countable subadditivity left out in class. Given a measure space $(\Omega, \mathcal{F}, \mu)$ and a sequence $\{A_n\}_n$ in $\mathcal{F}$, work out the following steps.

(i) Define another sequence $\{B_n\}_n$ in $\mathcal{F}$ by $B_1 := A_1$ and $B_n := A_n \setminus (A_1 \cup \cdots \cup A_{n-1}), \forall n > 1$. Show that $\bigcup_{n=1}^{m} A_n = \bigcup_{n=1}^{m} B_n$ for all m and $\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} B_n$.
(ii) Show that $\mu\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu(A_n)$. (Hint: use part (i))

Note: For probability measures, this inequality is usually referred to as the Boole's inequality.

Question 7. (Bonferroni's inequality for two events) Let A, B be two events in a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Show that

$$\mathbb{P}(A \cap B) \geq \max\{0, \mathbb{P}(A) + \mathbb{P}(B) - 1\}.$$

Can you generalize this to any finite number of events?

Question 8. (Inclusion-Exclusion principle for finitely many sets) Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $A_1, \ldots, A_n$ be events. Show that

$$\mathbb{P}(\bigcup_{i=1}^{n} A_i) = S_{1,n} - S_{2,n} + S_{3,n} - \cdots + (-1)^{n-1} S_{n,n},$$

where

$$S_{k,n} := \sum_{1 \le i_1 < i_2 < \ldots < i_k \le n} \mathbb{P}(A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}).$$

Question 9. Fix $x \in \mathbb{R}$. Verify

$$\left(-\infty, x + \frac{1}{n}\right] \downarrow (-\infty, x], \qquad \left(-\infty, x - \frac{1}{n}\right] \uparrow (-\infty, x)$$

as $n \to \infty$.